Launching the idea of data centers in space

Science X



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Les entreprises de technologie font flotter l'idée de construire des centres de données dans l'espace et d'exploiter l'énergie du soleil pour répondre aux exigences de puissance hors du monde dans une course féroce à l'intelligence artificielle.

La décision d'Elon Musk de faire en sorte que sa société de fusée SpaceX prenne en charge sa tenue d'intelligence artificielle xAI a ajouté du carburant au débat sur la faisabilité ou l'insensé des centres de données en orbite.

Qui sont les joueurs ?

SpaceX a donné le rythme sur le marché des lancements de fusées et Musk a parlé de mettre des centres de données dans l'espace.

La société de voitures électriques Tesla, dirigée par Musk, travaille également sur des robots humanoïdes, fournissant des travailleurs potentiels et des équipes de maintenance dans l'espace.

La start-up américaine Starcloud, à la fin de l'année dernière, a envoyé un satellite de la taille d'un réfrigérateur contenant une unité de traitement graphique Nvidia en orbite dans ce que le fabricant de puces d'IA a présenté comme un "début cosmique" pour le mini-centre de données.

Meanwhile, tech giant Google has laid out plans to launch test satellites by early 2027 as part of its Suncatcher project to build solar-powered data centers in space.

Blue Origin, the rocket and satellite company established by Amazon founder Jeff Bezos, is touting a TeraWave space-

based high-speed network that can be used by data centers to move information anywhere around the planet.

More than a dozen startups, aerospace leaders, and major tech firms are involved in the development, testing, or planning of space-based data centers.

Why look up?

The big draw of space for data centers is power supply, with the option of synchronizing satellites to the sun's orbit to ensure constant light beaming onto solar panels.

Building in space also avoids the challenges of acquiring land and meeting local regulations or community resistance to projects.

And advocates argue that data centers operating in space would be less harmful overall to the environment, aside from the pollution generated by rocket launches.

"The idea is that it will soon make much more sense to build data centers in space than it does to build them on Earth," Starcloud chief executive Philip Johnston said at a tech conference last year.

Current projects envision relying on clusters of low Earth orbit satellites positioned close enough together to ensure reliable wireless connectivity.

Lasers will connect space computers to terrestrial systems.

What could go wrong?

An obstacle to deploying servers in space has been the cost of getting them into orbit.

But a reusable SpaceX mega-rocket called Starship with massive payload potential promises to slash launch expenses.

Critical technical aspects of such operations need to be resolved, however, particularly damage to the orbiting data centers from high levels of radiation and extreme temperatures, and the danger of them being hit by space junk.

Another question is how malfunctioning or damaged gear would get fixed in an economical way.

Phillip Metzger, a professor in the University of Florida physics department and a former NASA scientist, reasoned in a recent online post that orbiting data center maintenance could be managed in ways such as using robots and small modular parts that could easily be replaced.

"A lot of the skepticism of data centers in space probably comes from failing to price in the effects of exponential expansion," Metzger said in a recent post on X, formerly Twitter.

"If AI doesn't grow exponentially, then I don't think it will make sense very soon to put them in space; but I think it will grow exponentially."

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Citation: Launching the idea of data centers in space (2026, February 4) retrieved 31 March 2026 from https://phys.org/news/2026-02-idea-centers-space.html

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